

Muñoz Rodríguez Iván

PhD



Iván Muñoz Rodríguez



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PROFESSIONAL PROFILE

Data-driven physicist with a PhD in Astrophysics and postdoctoral experience at CEA Paris-Saclay. Expert in analysing large, complex datasets using Python, statistical modelling, and simulation-based approaches. Proven track record in building end-to-end data pipelines, developing detection and classification algorithms, and working with real observational and geospatial data. Experienced in international, cross-disciplinary environments and motivated to apply rigorous quantitative methods to real-world, high-impact problems.

WORK EXPERIENCE

FRENCH ALTERNATIVE ENERGIES AND ATOMIC ENERGY COMMISSION (CEA) | POSTDOC

January 2025 – Today | Paris (France)

- Designed and maintained end-to-end data analysis pipelines for large X-ray observational datasets using Python.
- Developed and validated simulation-based models to test detection performance and systematic effects.
- Worked with large, non-modular legacy codebases; refactored and improved usability and reproducibility.
- Collaborated with international teams to deliver production-level catalogues used in large scientific consortia.

OMEGALAMBDATEC (OLT) | DATA SCIENTIST (INTERNSHIP)

May 2022 – July 2022 (3 months) | Garching, Munich (Germany)

- Translated raw satellite imagery into structured datasets suitable for downstream modelling and analysis.

PREWARP | DATA SCIENTIST (INTERNSHIP)

January 2021 – March 2021 (3 months) | Remote

- Contributed to demand forecasting and price optimisation models using regression techniques on real sales data.

CENTRO ASTRONÓMICO DE TIEDRA | SUPPORT ASTRONOMER

Summers 2019-2021-2022 | Tiedra, Valladolid (Spain)

- Delivered live scientific presentations to large, non-technical audiences under real-time operational constraints.
- Operated and troubleshooted technical observation and planetarium equipment during public events.
- Coordinated with small teams to ensure safe, reliable execution of high-attendance sessions.

EDUCATION

PHD IN ASTRONOMY

October 2020 – August 2024 | National Observatory of Athens (Greece) and University of Southampton (United Kingdom)

- Focus on large-scale data analysis, statistical inference, and model validation using simulations and observations.

MSC IN ASTRONOMY

2019-2020 | University of Granada (Spain)

BSC IN PHYSICS

2014 – 2019 | University of Salamanca (Spain)

TECHNICAL SKILLS

PROGRAMING & TOOLS

- Python (Pandas, NumPy, SciPy, Matplotlib, Astropy)
- Git, Bash, LaTeX
- Google Earth Engine (basic experience)
- Linux, Windows, macOS

DATA & ANALYSIS

- Large-scale data wrangling and statistical analysis
- Simulation-based modelling and validation
- Regression models and uncertainty analysis

SOFT SKILLS

- Analytical thinking: Comfortable reasoning under uncertainty and working with incomplete or noisy data.
- Problem ownership: Used to taking problems from vague requirements to validated solutions.
- Clear communication: Able to explain complex technical results to both technical and non-technical stakeholders.
- Collaboration: Extensive experience working in international, cross-disciplinary teams.
- Adaptability: Effective in fast-changing environments, from research pipelines to applied industry projects.
- Languages: Spanish (native), English (B2)

DEVELOPED SOFTWARE

- Developed simulation-based pipelines to generate realistic populations of supermassive black holes by statistically sampling observational constraints.
- Built 3D-to-2D projection tools for cosmological simulations, handling boundary conditions and volume stacking.
- Implemented satellite image preprocessing pipelines (Sentinel-2) including cloud masking, temporal averaging, and vegetation index extraction.
- Contributed to regression-based pricing and demand prediction modules in a commercial setting.